

# A Dense Incomplete Subspace with No Essential Schauder Basis

A candidate counterexample to both Problems 1 and 2 of arXiv:1806.07943

Automated mathematical discovery run `fa_banach_001`, lane 15

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**Status.** Candidate full counterexample, likely valid, requiring expert review. The bounded search described below found no explicit later resolution.

## Abstract

We answer both open problems of Coelho–Ribeiro–Salgado negatively. Let  $B$  be a separable Banach space without a Schauder basis and let  $X$  be the algebraic span of a countable dense subset of  $B$ . Then  $X$  is a separable, proper, dense, incomplete normed subspace with completion  $B$ . The source paper’s own Banach–Grunblum criterion says that an essential Schauder basis of  $X$  would extend to a Schauder basis of  $B$ , which is impossible. Since an essential unconditional basis is, by definition, an essential Schauder basis, the same  $X$  also answers the second problem negatively.

## 1 The two source problems

Coelho, Ribeiro, and Salgado ask whether every separable incomplete normed space admits an essential Schauder basis, and whether every incomplete normed space admits an essential unconditional basis [1].

## 2 Definitions and the completion mechanism

For a Schauder basis  $(x_n)$  of a normed space  $X$ , the source calls it *essential* when the synthesis map from its convergent coefficient space, equipped with the supremum norm of partial sums, onto  $X$  is an isomorphism. Its Theorem A states, in particular, that an essential basic sequence in  $X$  is an essential Schauder basis of the completion of its closed linear span in the completion  $\widehat{X}$  [1].

Consequently, if  $(x_n)$  is an essential Schauder basis of all of  $X$ , then the same sequence is a Schauder basis of  $\widehat{X}$ . The word “essential” adds nothing in the Banach space  $\widehat{X}$ , since every Schauder basis of a Banach space is essential in the source’s terminology.

## 3 Proof intuition

The obstruction to a basis can be placed entirely in the completion. Start with a separable Banach space  $B$  known to have no Schauder basis, but retain only the algebraic span of a countable dense set. This produces an incomplete space satisfying the source hypotheses without changing its completion. Essentiality is exactly strong enough, by the source’s Theorem A, to transport any proposed basis back to that forbidden completion.

number  $s \in \mathbb{N} \setminus \{0\}$  such that for all  $x \in N$ ,  $T(x) = \sum_{i=1}^s x_i \lambda_i \langle x, x_i \rangle$

where  $(\lambda_i)'$ s are the eigenvalues of  $T$ .

### 3. PROBLEMS

It is known that every Banach space with Schauder basis is separable (see [13, Proposition 4.1.10]). Banach has questioned if the converse holds: Does every separable Banach space admits a Schauder basis? In [9], Enflo gave a negative answer to this question, exhibiting a separable Banach space that has no Schauder basis.

**Problem 1.** *(The basis problem for normed spaces) Let  $X$  be a separable normed space which is not a Banach one. Does  $X$  admits an essential Schauder basis?*

In the sequence, we introduce the definition of (essential) unconditional basis.

**Definition 7.** Let  $X$  be a Banach space, and  $(x_n)_{n=1}^\infty$  a Schauder basis in  $X$ . The basis  $(x_n)_{n=1}^\infty$  is an unconditional basis if, for each  $x$  in  $X$ , there exists a unique expansion of the form

$$x = \sum_{n=1}^{\infty} a_n x_n$$

where the sum converges unconditionally.

Gowers and Maurey in [11] showed that there exists a Banach space that do not contain unconditional basis.

**Definition 8.** Let  $X$  be a normed space, and  $(x_n)_{n=1}^\infty$  an essential Schauder basis in  $X$ . The basis  $(x_n)_{n=1}^\infty$  is an essential unconditional basis if, for each  $x$  in  $X$ , there exists a unique expansion of the form

$$x = \sum_{n=1}^{\infty} a_n x_n$$

Figure 1: Problem 1 and its immediate context in the source, PDF p. 5.

The unconditional question needs no separate convergence argument. Under the source's Definition 8, "essential unconditional" is an additional property of an already essential Schauder basis. The first obstruction therefore settles both questions.

## 4 Counterexample

**Theorem 1.** *There exists a separable incomplete normed space  $X$  which has neither an essential Schauder basis nor an essential unconditional basis. Hence Problems 1 and 2 of [1] both have negative answers.*

*Proof.* Let  $B$  be a separable infinite-dimensional Banach space without a Schauder basis. Such a space exists by Enflo's construction of a separable Banach space without the approximation property [2], since a Schauder basis gives bounded finite-rank partial-sum projections converging pointwise to the identity and hence the bounded approximation property.

Choose a countable dense set  $D = \{d_1, d_2, \dots\}$  in  $B$  and put

$$X = \text{span } D$$

with the norm inherited from  $B$ . The space  $X$  is separable and dense in  $B$ . It is proper: otherwise  $B$  would have countable Hamel dimension, hence would be the union of the increasing finite-dimensional closed subspaces  $\text{span}\{d_1, \dots, d_m\}$ . This contradicts the Baire category theorem because  $B$  is infinite-dimensional. Since a complete subspace of a Banach space is closed, the proper dense subspace  $X$  is incomplete. Its completion is canonically isometric to  $B$ .

We end this section with the following question.

**Problem 2.** *(The unconditional basic sequence problem for normed spaces) Let  $X$  be a normed space which it is not a Banach space. Does  $X$  admits an essential unconditional basis?*

2.1. **Organization of the text.** In Sections 1, 2 and 3 we provide preliminary definitions

Figure 2: Problem 2 in the source, PDF p. 6.

Suppose that  $(x_n)$  were an essential Schauder basis of  $X$ . Then  $\overline{\text{span}}^X\{x_n : n \geq 1\} = X$ . Applying item (ii) $\Rightarrow$ (i) of Theorem A in [1],  $(x_n)$  is an essential Schauder basis of the completion of this closed span, namely  $B$ . Thus  $B$  has a Schauder basis, contrary to its choice. Therefore  $X$  has no essential Schauder basis.

Finally, Definition 8 of [1] defines an essential unconditional basis to be an essential Schauder basis whose expansions converge unconditionally. Since  $X$  has no essential Schauder basis at all, it has no essential unconditional basis. This same  $X$  disproves both universal statements.  $\square$

## 5 Verification, limitations, and novelty

The verification report checks separately: properness by Baire category, incompleteness and identification of the completion, the exact application of Theorem A, and the definitional implication for unconditional bases. No finite computation is relevant.

The result is existential because it inherits the known Enflo space; it does not provide a simpler concrete norm formula for  $B$ . It also does not classify which incomplete normed spaces do possess essential bases.

A bounded search on 11 August 2026 covered the run’s registry, solution, attempt, and active-claim indexes; the exact arXiv id and title; both exact problem phrases; the November 2024 arXiv/published version; and combinations of dense subspaces, completions, Enflo spaces, and essential Schauder bases. The current version still states both problems, and no explicit answer was found. This is provisional novelty evidence only.

## Human-review recommendation

Check first that Theorem A of the source applies to a basis spanning all of  $X$ , so its completed closed span is precisely  $\widehat{X} = B$ . Then verify that Definition 8 makes the second conclusion immediate. The Baire-category and completion steps are standard.

## References

- [1] V. Coelho, J. Ribeiro, and L. Salgado, *A note on Basis Problem in normed spaces*, arXiv:1806.07943, Problems 1–2 and Theorem A; published version, 2024.
- [2] P. Enflo, *A counterexample to the approximation problem in Banach spaces*, Acta Math. **130** (1973), 309–317.